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Research on optimizing turbo-matching of a large-displacement PFI hydrogen engine to achieve high-power performance

Feng-yu Lai [a], *Bai-gang Sun* [a,b], *Gang Xiao* [c], *Qing-he Luo* [a,b], *Ling-zhi Bao* [a,*]

[a] *School of Mechanical Engineering, Beijing Institute of Technology, Beijing 100081, China*
[b] *Beijing Institute of Technology Chongqing Innovation Center, Chongqing, 401121, China*
[c] *Guangxi Yuchai Machinery Co., Ltd., Guangxi 537005, China*

HIGHLIGHTS

- The CNG engine turbine is not suited to H2ICE due to the low reduced mass flow rate.
- The method considers exhaust components and temperature to reach desired power.
- The max power is improved from 93.0 kW to 128.9 kW with the optimized turbine.
- The output torque is reduced by 15.6% at 1200 rpm due to poor intake pressure.
- The feature of turbocharging of H2ICE is discussed with some potential solutions.

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ABSTRACT

The output power of hydrogen engines is restricted due to the low volume energy density of hydrogen and lean burn, thus a turbocharger is necessary to boost the power intensity, especially for large-displacement hydrogen engines. However, turbine selection for hydrogen engines is complex comparing gasoline engines because of the different exhaust components and low exhaust temperature. In this research, the turbocharger-matching of a 5.13 L hydrogen engine is studied, and the results show the original turbine, whose maximum reduced mass flow is 0.00966 (kg/s)·K^0.5/kPa, cannot achieve the desired power performance at high speeds. The reduced mass flow rate can be calculated, taking into account the effect that exhausts composition and temperature. The accuracy of the method is validated using data from a 2.3 L turbocharged H2ICE at various engine speeds and throttle openings, and the error between the measured and calculated expansion ratio is less than 5% mostly. At last, a well-matching turbine for high engine speeds of the 5.13 L hydrogen engine to meet the desired power of 120 kW, whose maximum reduced mass flow is extended to 0.0156 (kg/s)·K^0.5/kPa, is selected according to the method, and the maximum power reaches 128.9 kW with a maximum torque of 662.6 N m, improving by 38.7% and 7.7% respectively comparing with the original turbine. However, the output torque reduces by 15.6% at 1200 rpm. The reduced mass flow rate varies significantly in comparison to the turbines that are well suited for low and high engine speeds

* *Corresponding author*. School of Mechanical Engineering, Beijing Institute of Technology, Beijing 100081, China.
E-mail addresses: xiaogang@yuchai.com (G. Xiao), 783572692@bit.edu.cn (L.-z. Bao).
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respectively, thus the Variable Geometry Turbine may be the solution for both low and high speeds simultaneously for hydrogen engines.

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Nomenclature

H2ICE	hydrogen internal combustion engine
HFC	hydrogen fuel cell
PFI	port fuel injection
DI	direct injection
VGT	variable geometry turbine
BTE	brake thermal efficiency
BMEP	break mean effective pressure
MBT	minimum spark advance for best torque
$\dot{m}_H$	hydrogen mass flow rate
$\dot{m}_C$	air mass flow rate
$\dot{m}_L$	air mass flow through the compressor
$\dot{m}_{reduced}$	reduced mass flow rate of the turbine
$\dot{m}_{water}$	cooling water mass flow rate
T_1	assumed exhaust gas temperature
T_2	theoretical turbine inlet temperature
$T_{water-in}$	cooling water inlet temperature
$T_{water-out}$	cooling water outlet temperature
T_C	air temperature after the intercooler
P_e	desired power
η	desired brake thermal efficiency
λ	excess air ratio
H_u	hydrogen heat value
L_0	stoichiometric air-fuel ratio
ω	mass fraction
$\eta_{cooling}$	the cooling to fuel power ratio
η_{TL}	turbocharger efficiency
$C_{p\ mix}$	specific heat capacity at constant pressure of the exhaust
$C_{p\ intake}$	specific heat capacity at constant pressure of the inlet air
$C_{v\ mix}$	specific heat capacity at constant volume of exhaust
$C_{p\ water}$	specific heat capacity at constant pressure of cooling water
$p_{turbine\ discharge}$	turbine discharge pressure
$p_{pre-turbine}$	pre-turbine exhaust pressure
κT	gas adiabatic index in the turbine
κL	gas adiabatic index in the compressor

Introduction

Due to the lack of fossil fuels and the global greenhouse impact, researchers are motivated to study and develop alternative fuels such as hydrogen and ammonia. Hydrogen is appealing as a prospective energy carrier. Hydrogen internal combustion engines (H2ICEs) and hydrogen fuel cells (HFCs) are known as the two main ways to utilize hydrogen energy, while H2ICEs have a higher power density and high efficiency at part load [1–3] compared with HFCs. Many benefits of using hydrogen as a fuel in ICEs include reduced CO2 emissions, rapid combustion velocity, low ignition energy, and high thermal efficiency [1,4].

Nevertheless, owing to hydrogen's low volume energy density and lean burn, which aims to produce almost near-zero NOx emissions [5], H2ICEs have significantly less power than gasoline engines. There are two main types of H2ICEs mostly being investigated, port fuel injection (PFI) and direct injection (DI). PFI H2ICEs are more easily and cheaply transformed from gasoline engines than DI H2ICEs [6]. However, the increase in power is limited by anomalous combustion (preignition, backfire, and so on) [7,8]. According to the study of Karagöz et al. [9], the lower low heat value in volume than that of gasoline, as well as backfire, restricts power output increase. Although DI H2ICEs are more difficult to implement due to the complexity of injector lubrication and cooling, as well as the large cycle variation [10], DI can increase air flow and DI has been found an effective way to boost power output and lower NOx emissions [11]. However, there is still a power performance disparity between H2ICEs and gasoline engines. Supercharging and turbocharging are adopted by researchers to increase the power intensity of H2ICEs, and the results indicate that they are beneficial to improving power output and thermal efficiency. Gürbüz et al. [12] conducted supercharging tests on a 0.476 L hydrogen engine with an external supercharging air blower. The results demonstrated a 38% increase in indicated power and a 16.5% rise in fuel consumption respectively when the equivalence ratio is 0.6, proving the potential to improve the performance of the turbocharged H2ICE. Besides, the researchers [13] revealed that intake-air temperature has obvious effects on the flame development angle and volumetric efficiency. Ford reviewed the development process of a 6.8 L supercharged V-10 H2ICE [14], and power performance was comparable to or better than that of target CNG or gasoline engines. Nakajima et al. [15] demonstrated that turbocharging boosted brake thermal efficiency by 2% at λ (excess air ratio) about 2.5 for a PFI 1.3 L H2ICE. Lohse-Busch et al. [16] reported that a 170 N m torque was available for a 2.3 L PFI H2ICE at 3000 rpm with a λ of 2 because supercharging improved volume efficiency. Takagi et al. [17] found that supercharging increased IMEP of a 1 L one-cylinder DI H2ICE from 0.6 MPa to 0.85 MPa while keeping λ at 1.8. Furthermore, thermal efficiency was improved to nearly 50%, which attributed to the reduced proportion of the cooling loss due to the increased power output. The poor improvement in power output at high engine speeds caused by the energy consumption limits the implement of superchargers, whereas turbochargers are more common in automobiles because of the cheaper cost. The results of Wallner et al. [18] and Obermair et al. [19] also revealed turbocharging improved the indicated thermal

efficiency of a 0.66 L DI H2ICE from 43% to 45.6% at 2000 rpm with an IMEP of 0.8 MPa. Jilakara et al. [20] reported that the power output of a 2.3 L turbocharged H2ICE reached 66.7 kW at 3600 rpm while a naturally aspirated CNG engine reached 52 kW. The research of Bao et al. [21] showed the power of a matched turbocharged hydrogen engine increased by 123%, and the torque increased by 195% comparing naturally aspirated engine.

However, there are still some problems to be resolved in turbocharging of H2ICE. First, the H2ICE turbo cannot sustain the energy necessary to provide compressor required boosting because of low exhaust temperature at low engine speeds [20]. Secondly, increased back pressure and pumping loss restrict the enhancement of power output, particularly under high engine speed condition [22]. According to the research of Lee et al. [23], the back pressure of a turbocharged 2.4 L PFI H2ICE reached 0.231 MPa while intake pressure was 0.132 MPa at 6000 rpm, and the torque of 84 N m was lower than that of 98 N m under naturally aspirated condition. Therefore, choosing a well-matched turbocharger for H2ICE is significant to boost the power output further. A more powerful compressor with greater air mass flow rates and higher-pressure ratio is better suited for H2ICE [21], and Luo et al. [24] suggested a method for selecting a compressor considering the gas partial pressure of hydrogen in cylinders in previous work. Meanwhile, there is still a necessity to select a well-matched turbine. Bao et al. [21] stated that a small turbo can provide sufficient power for turbocharging, however, the reduction of turbine diameter causes a rise of exhaust pressure especially at high speeds. Similar conclusions can be summarized in other researches, but the turbocharging of H2ICE is studied through exhaust pressure, and H2ICE turbine have not been quantitatively analyzed with reduced mass flow rates.

In this research, the turbocharger-matching of a 5.13 L hydrogen engine is studied, and the results show the original turbine cannot achieve the desired power performance at high speeds. A method is proposed based on former researches in order to select a turbine for a H2ICE taking the exhaust component into consideration, and the reduced mass flow rate can be calculated. Experimental data were used to validate the accuracy of this method. Finally, a well-matching turbine for high engine speeds of the large-displacement PFI hydrogen engine to provide the desired power is selected according to the method, and characteristics of the large-displacement PFI hydrogen engine turbocharger-matching are concluded by comparing the data at low and high speeds.

Table 1 – The specifications of the large-displacement PFI engine.

Items	Unit	Values
Number of cylinders		4
Bore	mm	110
Stroke	mm	135
Displacement	L	5.132
Number of valves		4
Compress ratio		12.5
EVO	CA	47.7°bBDC
EVC	CA	9.4°aTDC
IVO	CA	−2.4°bTDC
IVC	CA	15°aBDC
Maximum power	kW	176
Maximum torque	N·m	900
Revolution of the engine	rpm	600–2000

Materials and methods

The engine test system and data acquisition system were properly arranged and connected in order to obtain accurate and precise experimental data. The 5.13 L 4-cylinder PFI turbocharged hydrogen engine was transformed from a CNG engine. The specifications of the large-displacement PFI hydrogen engine are shown in Table 1 and the layout of the experiment and data acquisition systems are shown in Fig. 1.

A CW250 eddy current dynamometer, an oil and water temperature controller, air conditioning, and a ToCeiL20 N hot-film air flow meter were all included in the test system. The CW250 dynamometer has a maximum working capacity of 250 kW and a maximum speed of 8000 rpm. The temperature of the oil and cooling water in the engine were controlled

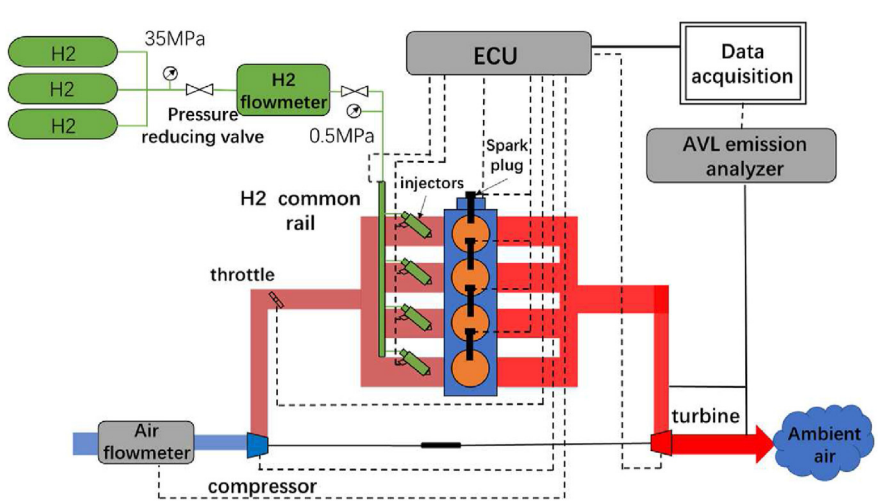


Fig. 1 – Layout of the 5.13 L PFI hydrogen engine experiment and data acquisition systems.

Table 2 – The uncertainties of the measured parameters.

Parameters	Uncertainties	Range
Speed	± 0.1%	0–10000 rpm
Torque	± 0.4%	0–1100 N m
Crank angel	± 0.02 °CA	
Pressure	± 0.8%	0–200 MPa
Air mass flow	± 1% Full Scale	0–450 kg/h
Hydrogen mass flow	± 0.1% Full Scale	0–6.5 kg/h

by oil and water temperature controllers. The air mass flow into the engine was measured with the ToCeiL20 N hot-film air flow meter, and an Emerson CMF025 coriolis mass flow meter was used to determine the hydrogen mass flow. The Kibox combustion analyzer was used in the data acquisition system, and the cylinder pressure and crankshaft angle signals were recorded using a Kistler 6117B sensor and a Kistler 2613B sensor, respectively. The specifications of uncertainties of the measured parameters are listed in Table 2. The engine was warmed up before the testing to ensure that the oil and cooling water reached 90 °C and 80 °C, respectively. The tests were carried out while the engine was running at a constant speed, with intervals of 500 rpm between 1500 and 4000 rpm. The tests were carried out at the MBT (minimum spark advance for best torque) in order to obtain comparable data, and MBT tests were conducted by modifying the ignition timing under different engine speed and load conditions. After the hydrogen engine worked steadily at the current working condition for 2 min, the average value of the test data of 200 cycles at this working condition was measured and recorded. The uncertainty of tested results can be calculated according to the principle of root-mean square method to obtain the magnitude of the error given by Gaussian distribution [25], as shown in Formula (1):

$$\Delta R = \left[\left(\frac{\partial R}{\partial x_1}\Delta x_1\right)^2 + \left(\frac{\partial R}{\partial x_2}\Delta x_2\right)^2 + \ldots\left(\frac{\partial R}{\partial x_n}\Delta x_n\right)^2\right]^{1/2} \quad (1)$$

where ΔR is the uncertainty in the computed result; R is a given function of the computed results; x_1, x_2, and x_n are the different measured parameters; Δx_1, Δx_2, Δx_n are the corresponding uncertainty of the different measured parameters. The test engine was a 2.3 L four-cylinder PFI turbocharged H2ICE, and its experiment data were used to confirm the method's predicted results. The specifications of the experimental hydrogen engine are shown in Table 3.

Table 3 – The specifications of test engine.

Items	Unit	Values
NO. of cylinders		4
Bore	mm	90
Stroke	mm	90
Displacement	L	2.299
NO. of valves		4
Compress ratio		9.3
EVO	CA	53.64°bBDC
EVC	CA	44.36°aTDC
IVO	CA	23.3°bTDC
IVC	CA	58°aBDC

Results and discussion

Original turbocharger-matching of large-displacement PFI hydrogen engine

The desired power performance is set at 600 N m @ 1200 rpm with a maximum power of 120 kW at high speed. The experiments are conducted with the original turbocharger which is suited with the original CNG engine for preliminary performance exploration. In order to achieve the desired torque, the intake pressure increases to 0.312 MPa and the air flow reaches 448 kg/h. The results show the output torque can achieve 615 N m with a lambda of 2.12 and air flow rate of 448 kg/h at 1200 rpm. In addition, the expansion ratios surpass 3.0 when H2ICE is operated at high load, and the maximum reaches 3.43. The power performance is acceptable at 1200 rpm. However, the power is limited to 93 kW with a lambda of 1.77 at 1800 rpm, while the increasing hydrogen mass flow is restricted by the knock phenomena and air mass flow has reached its maximum at 498.6 kg/h. The reduced mass flow rate of the turbine can be calculated and analyzed with the efficiency map based on the pre-turbine exhaust pressure and temperature with the hydrogen and air mass flow recorded. Fig. 2 (a) shows the expansion ratios at 1200 rpm remain high, and the reduced mass flow rate at 1800 rpm is higher than the boundary data of the turbine efficiency map, which shows poor matching and leads to restricted output power. A series of experiments at speeds of 2000 rpm and 1600 rpm have been carried out in an effort to increase the maximum power and study the turbocharger matching at the current speed. The results of Fig. 2 (a) and Fig. 2 (b) demonstrate that the reduced mass flow rates at 2000 rpm are much higher than the boundary data, but the BMEP at that speed is 10 bar which is far below the desired power performance. On the other hand, the output power at 1600 rpm is also tested as Fig. 2 (a) and Fig. 2 (b) shown. The lower reduced mass flow rate is acceptable for the original turbine when the hydrogen engine is operated at medium and low loads, however, the reduced mass flow rate is no longer within the turbine map when the BMEP rises to 10.98 bar as the air flow grows. The findings of these experiments show that the desired power performance at high speed cannot be achieved with the original turbine, requiring a well-matched turbine. Further research is aimed at the selection of turbine at high speed.

Analysis of the method to select a well-matched turbine for large-displacement PFI hydrogen engine

Commercial 1D codes are frequently used nowadays to match gasoline engine turbochargers, while the simulations need quantities of data especially cooling loss for H2ICE calibration. Due to the smaller quenching distance and higher heat value of hydrogen, some study on the heat transfer of hydrogen engines indicates that the heat transfer model is distinct from the uncalibrated Woschni model, which is typically chosen in gasoline engine models [26]. Researchers found inaccurate results when adopting unchanged models to hydrogen engines [27], and that leads to inaccurate results of the pre-turbine temperature. Thus, calibration for heat transfer of a

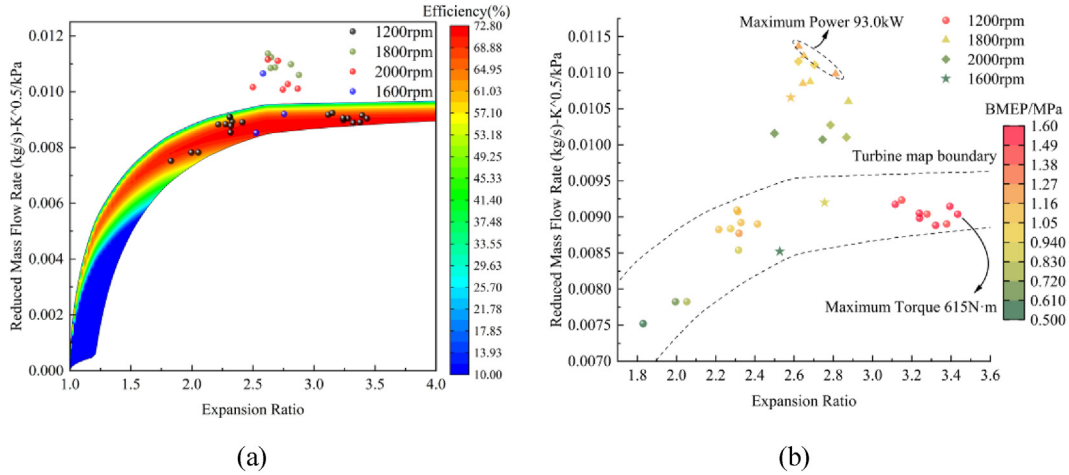


Fig. 2 – (a) The 5.13 L hydrogen engine matching points in the original turbine efficiency map and (b) BMEP variation at different conditions.

hydrogen engine based on the abundant energy distribution experiments results is necessary to ensure the accuracy of the 1D simulation and the turbocharger matching. On the other hand, a generally well-matching turbine can only be chosen after multiple computations of various turbochargers, as the 1D simulation is carried out depending on the map data input of the turbocharger. The experiment results have been carried out with the original turbocharger can serve as the basis for selecting a well-matched turbine. The results above show that the hydrogen engine's performance is significantly impacted by the reduced mass flow rate. Thus, calculated reduced mass flow rates at certain conditions can be a valuable reference for selecting a turbine.

A calculation method for turbine selection is proposed based on the calculation of Zinner et al. [28], and the reduced mass flow rate can be achieved according to the calculated turbine inlet temperature and the turbine expansion ratio. The following are the steps of this calculation method, and Fig. 3 shows the process of the calculation method.

1) The excess air ratio (λ) is specified, and the flow rates of hydrogen and air are calculated according to the designed power and the brake thermal efficiency (BTE) of H2ICE.
2) The mass fraction ω of each exhaust component is calculated according to the chemical equation for the reaction between hydrogen and air.
3) The specific heat capacity at a constant pressure of the exhaust gas mixture (*Cp_mix*) is calculated by assuming the exhaust gas temperature T_1 and querying the *Cp* corresponding to each exhaust gas component.
4) The theoretical value of the turbine inlet temperature T_2 is calculated according to the desired brake effective thermal efficiency η expected at the design point of the H2ICE and the cooling to fuel power ratio $\eta_{cooling}$, an empirical parameter, under the corresponding operating conditions. The pre-turbine exhaust temperature can be determined when T_2 is equal to T_1. The expansion ratio of the turbine can be calculated according to the turbocharger efficiency, the pre-turbine exhaust temperature, and the pressure ratio of the compressor.

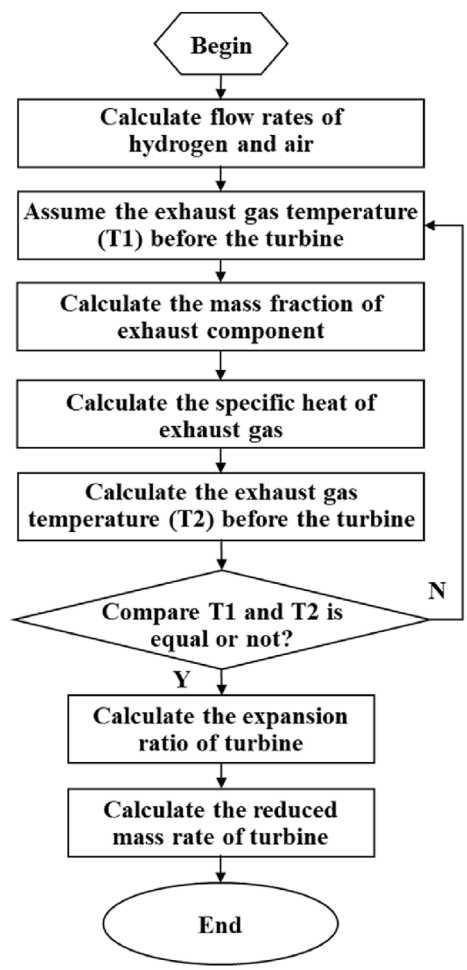


Fig. 3 – The process of the calculation method.

5) The pre-turbine exhaust pressure may be estimated using the expansion ratio with the turbine discharge pressure, which is related to the hydrogen engine's speed. The calculated reduced mass flow rate is obtained according to the parameters above.

According to the desired power and BTE of H2ICE, as well as the specified excess air ratio, the air mass flow and hydrogen mass flow can be determined by Formula (2) (3)

$$\dot{m}_H = \frac{P_e \times 1000}{H_u \eta} \tag{2}$$

$$\dot{m}_C = \lambda L_0 \dot{m}_H \tag{3}$$

where $\dot{m}_H$ is the hydrogen flow rate, kg/s; P_e is the desired power, kW; η is the brake thermal efficiency; H_u is the hydrogen heat value, kJ/kg; $\dot{m}_C$ is the air flow rate, kg/s; L_0 is the theoretical amount of air required for complete combustion per kilogram of hydrogen; λ is the specified excess air ratio;

Next, the mass fraction ω of the exhaust composition after the reaction can be calculated by Formula (4) (5) (6).

$$\omega_{H_2O} = \frac{2 \times 18}{2 \times 18 + (\lambda - 1) \times 32 + 3.71 \cdot \lambda \times 28} \tag{4}$$

$$\omega_{O_2} = \frac{(\lambda - 1) \times 32}{2 \times 18 + (\lambda - 1) \times 32 + 3.71 \cdot \lambda \times 28} \tag{5}$$

$$\omega_{N_2} = \frac{3.71 \times \lambda \times 28}{2 \times 18 + (\lambda - 1) \times 32 + 3.71 \cdot \lambda \times 28} \tag{6}$$

where ω_{H_2O} is the mass fraction water vapour; ω_{O_2} is the mass fraction oxygen; ω_{N_2} is the mass fraction nitrogen;

According to the research [29], the specific heat capacity in turbine affects the calculation of turbine. Thus, the specific heat capacity at constant pressure of the exhaust gas mixture (*Cp_mix*) is re-calculated by Formula (7). The exhaust gas temperature T_1 can be assumed, and the specific heat capacity at constant pressure *Cp* for each exhaust gas component can be searched (Temperature is the most important component impacting *Cp*, and the pressure has a less impact on *Cp*. Hence 0.25 MPa is chosen as the appropriate pressure.).

$$C_{p\ mix} = \omega_{H_2O} C_{p\ H_2O} + \omega_{O_2} C_{p\ O_2} + \omega_{N_2} C_{p\ N_2} \tag{7}$$

where: $C_{p\ mix}$ is the specific heat capacity for the exhaust gas mixture, kJ/kg·K; $C_{p\ H_2O}$ is the specific heat capacity for water vapour, kJ/kg·K; $C_{p\ O_2}$ is the specific heat capacity for oxygen, kJ/kg·K; $C_{p\ N_2}$ is the specific heat capacity for nitrogen, kJ/kg·K.

The fourth part is the calculation to determine the pre-turbine temperature and expansion ratio. The theoretical value of the turbine inlet temperature T_2 is calculated by Formula (8), based on the desired effective thermal efficiency η at the design point of the H2ICE, and the cooling to fuel power ratio $\eta_{cooling}$ under the operating conditions, which is an empirical parameter and needs to be set reasonably. At the early development of the hydrogen engine, the certain value of $\eta_{cooling}$ is based on the cooling data of the hydrogen engine with similar displacement or, if available, preliminary experiments under certain conditions.

$$T_2 = \frac{(1 - \eta - \eta_{cooling}) H_u + \lambda L_0 C_{p\ intake} T_C}{(1 + \lambda L_0) C_{p\ mix}} \tag{8}$$

where: $\eta_{cooling}$ is the cooling to fuel power ratio; $C_{p\ intake}$ is the specific heat capacity at constant pressure of the inlet air, kJ/kg·K; and T_C is the temperature of the air after the intercooler, K.

The temperature T_1 is determined as the pre-turbine exhaust temperature when the difference between the assumed values T_1 and calculated values T_2 is within the permissible error range. This method improves the accuracy of the calculation of the pre-turbine exhaust temperature and the turbine expansion ratio. Formula (9) can be obtained from the research of Zinner et al. [28], and the pre-turbine exhaust pressure can be obtained from Formula (10). At last, the turbine reduced mass flow rate at certain condition can be calculated by in Formula (11).

$$\pi_T = \frac{1}{\left\{1 + \left[1 - (\pi_C)^{\frac{\kappa L - 1}{\kappa L}}\right] \cdot \frac{\dot{m}_L}{\dot{m}_T} \cdot \frac{C_{p\ intake}}{C_{p\ mix}} \cdot \frac{T_C}{T_1} \cdot \frac{1}{\eta_{TL}}\right\}^{\frac{\kappa T}{\kappa T - 1}}} \tag{9}$$

$$p_{pre-turbine} = p_{turbine\ discharge} \cdot \pi_T \tag{10}$$

$$\dot{m}_{reduced} = \frac{(\dot{m}_H + \dot{m}_C) \cdot \sqrt{T_1}}{p_{pre-turbine}} \tag{11}$$

where: $\dot{m}_L$ is the air flow through the compressor, kg/s; κL is the gas adiabatic index in the compressor; $\dot{m}_T$ is the gas flow through the turbine, kg/s; κT is the gas adiabatic index in the turbine, which is also can be determined by the calculated $C_{p\ mix}$ and $C_{v\ mix}$; π_T is expansion ratio of the turbine; π_C is pressure ratio of the compressor; η_{TL} is the efficiency of the turbocharger; $p_{pre-turbine}$ is the pre-turbine exhaust pressure, MPa; $p_{turbine\ discharge}$ is the turbine discharge pressure, MPa; $\dot{m}_{reduced}$ is the reduced mass flow rate of the turbine, (kg/s)·K^0.5/kPa;

Validation of the method with the experimental data

To verify the accuracy of this method for selecting a turbine for the H2ICE, experimental data from a 2.3 L four-cylinder turbocharged H2ICE at 2000 rpm, 2500 rpm, 3000 rpm, and 4000 rpm are utilized to compare with the calculated values. The turbine inlet temperature, turbine inlet pressure and turbine outlet pressure in the experimental data are measured by the corresponding sensors. Formula (12), which incorporates cooling water inlet and outlet temperatures, flow rates, can be used to calculate the cooling to fuel power ratio $\eta_{cooling}$, and it can only be used to compute the cooling when the required sensors are installed in order to validate the measured and estimated turbine inlet temperature.

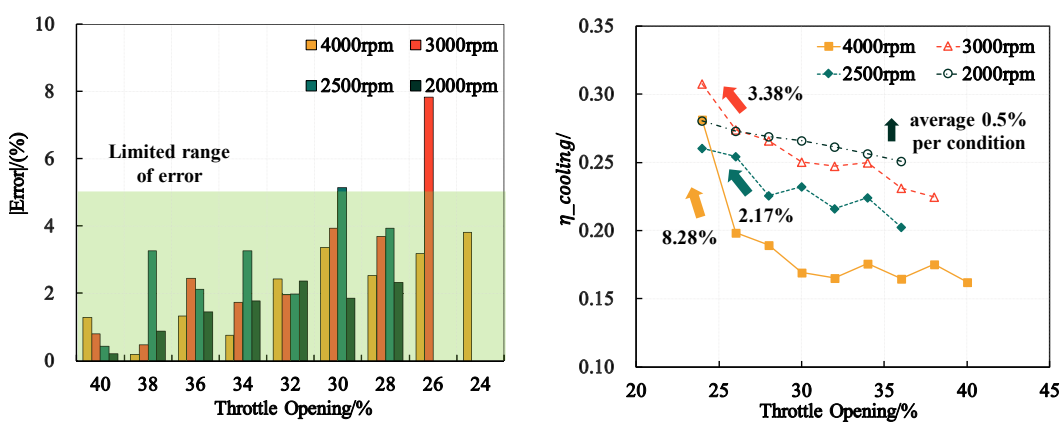
$$\eta_{cooling} = \frac{\dot{m}_{water} \cdot (T_{water-out} - T_{water-in}) \cdot C_{p\ water}}{\dot{m}_H \cdot H_u} \tag{12}$$

where: $\dot{m}_{water}$ is the flow rate of cooling water, kg/s; $T_{water-out}$ is outlet temperature of cooling water, K; $T_{water-in}$ is inlet temperature of cooling water, K; $C_{p\ water}$ is the specific heat capacity at constant pressure of cooling water, kJ/kg·K.

As the hydrogen flow rate is much lower than the air flow rate and the temperature of the injected hydrogen is about 300 K, the energy caused by the initial state of the hydrogen is negligible and the effect of the hydrogen calorific value is mainly considered. The results of turbine inlet temperature of the H2ICE can be calculated by Formula (8) and (12). The calculated and tested data of turbine inlet temperatures,

Table 4 – The calculated and measured data of turbine inlet temperatures (Kelvin).

Throttle	4000 rpm			3000 rpm		
	Calculated	Measured	|Error|/%	Calculated	Measured	|Error|/%
40%	943.1	931.15	1.28			
38%	934.8	936.55	0.19	889.1	882.2	0.79
36%	957.7	945.15	1.33	880.0	884.2	0.47
34%	948.8	941.75	0.75	854.4	875.8	2.44
32%	964.6	941.75	2.43	865.2	880.6	1.74
30%	970.8	939.15	3.38	861.3	878.6	1.96
28%	965.3	941.55	2.52	842.6	877.2	3.94
26%	965.2	935.35	3.19	846.3	878.8	3.69
24%	888.1	923.35	3.81	806.3	874.8	7.83
	2500 rpm			2000 rpm		
Throttle	Calculated	Measured	|Error|/%	Calculated	Measured	|Error|/%
36%	884.9	881.2	0.42	827.9	826.2	0.21
34%	861.2	890.2	3.26	819.6	827.0	0.88
32%	868.4	887.2	2.11	811.3	823.2	1.44
30%	855.5	884.4	3.26	806.9	821.6	1.78
28%	868.3	886.0	1.99	805.6	825.2	2.37
26%	838.8	884.4	5.15	802.0	817.2	1.85
24%	843.5	878.2	3.94	799.0	818.0	2.32



(a) Variation in error of turbine inlet temperature at different engine speeds

(b) Variation in $\eta_{cooling}$ at different engine speeds

Fig. 4 – Variation in $\eta_{cooling}$ and the error of turbine inlet temperature at different engine speeds.

when the equivalence ratio is 0.55 and the ignition timing is set at MBT, are shown in Table 4.

The Error can be calculated by the formula followed:

$$\text{Error} = \frac{\textit{Calculated Data} - \textit{Measured Data}}{\textit{Measured Data}} \tag{13}$$

According to Fig. 4 (a), the calculated values for the turbine inlet temperature have an error margin of less than 5% generally, which can be serve as the criterion in this method [24]. At 3000 rpm, however, the maximum error 7.83% occurs. This is most likely a result of measurement error, which led to a greater $\eta_{cooling}$ being calculated. When the throttle opening grows, the $\eta_{cooling}$ reduces, as shown by the findings in Fig. 4 (b). Because the varied throttle openings at each operational condition, as at 2000 rpm, are not significant, the $\eta_{cooling}$ should not fluctuate significantly. However, the $\eta_{cooling}$ fluctuates significantly for a few conditions where the error of predicted temperature is substantial, thus it is critical to set the $\eta_{cooling}$ accurately for each condition. Fig. 5 illustrates the comparison between the calculated expansion ratio and the measured expansion ratio at different speeds and throttle openings. The results reveal that as the throttle opening increases, the increasing airflow causes the expansion ratio to rise as well. As a result, as the compressor's pressure ratio rises, a greater expansion ratio may be achieved. However, the error in the expansion ratio calculation is substantially lower than that in the temperature calculation when comparing the results of Figs. 4 and 5. The hydrogen engine is operated at low load and has a low compressor pressure ratio, resulting in a reduction in the expansion ratio error transmitted by the temperature error.

Additionally, the experiments data of the 5.13 L H2ICE are also utilized to confirm the applicability of different displacements. According to Fig. 6 (a), the error between

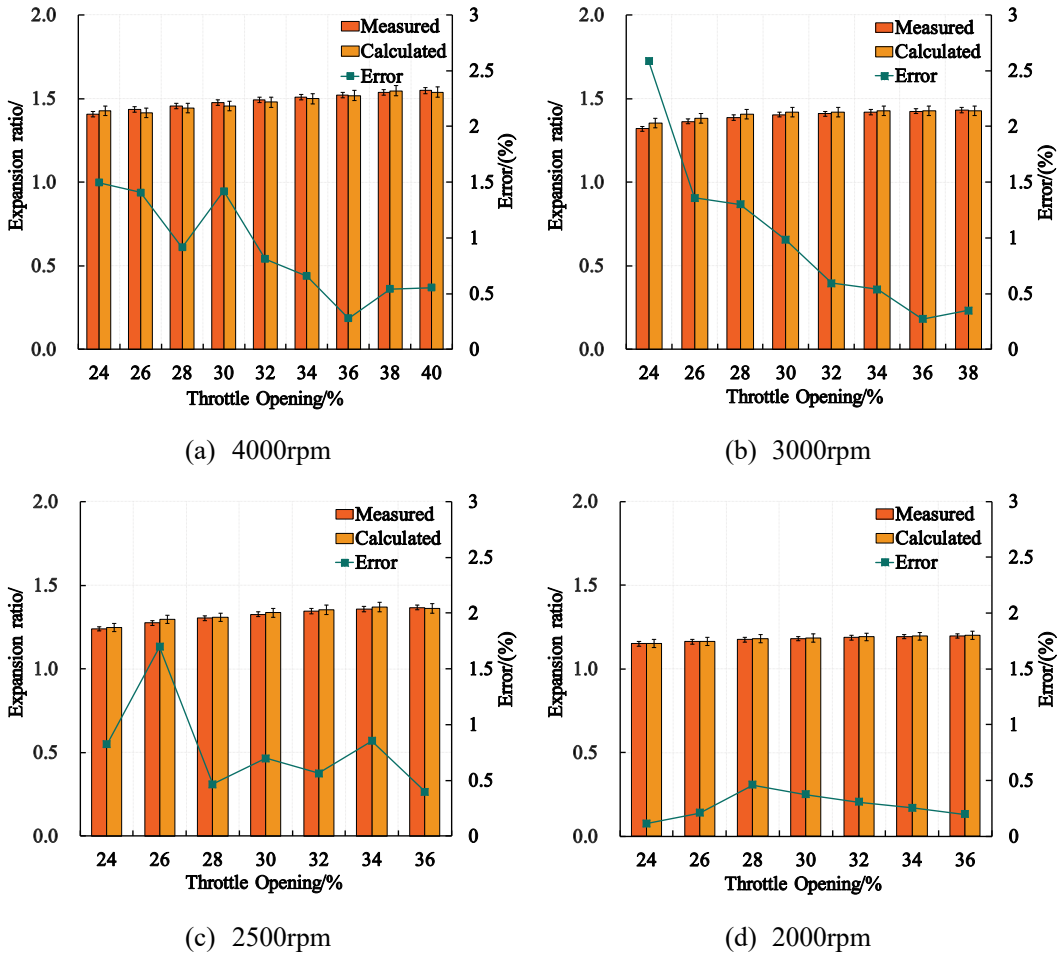


Fig. 5 – Comparison between the measured and calculated expansion ratio at different engine speeds.

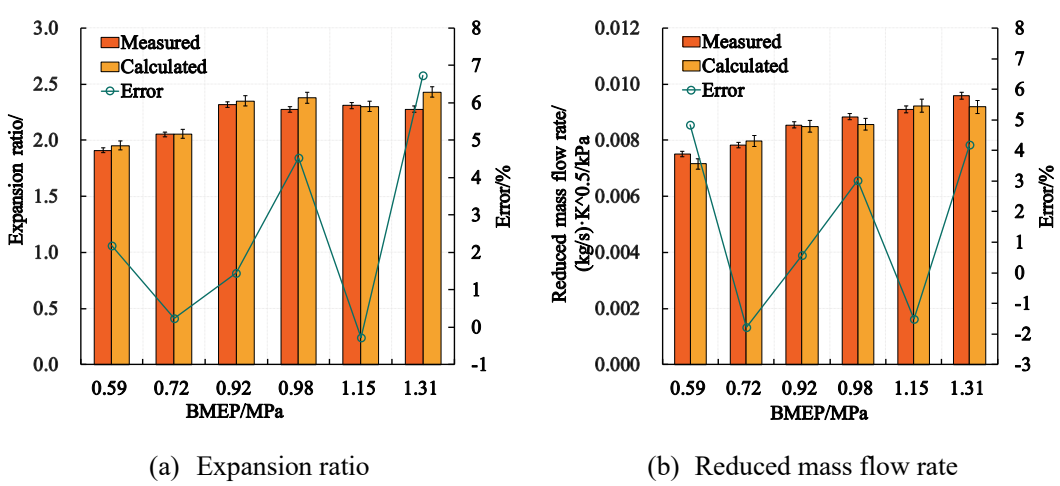


Fig. 6 – Comparison between the measured and calculated (a) expansion ratio and (b) reduced mass flow rate at 1200 rpm.

measured and calculated expansion ratio is generally within 5%, and overly wide waste gate opening can cause the inaccuracy. The difference between the calculated and measured reduced mass flow rate is shown in Fig. 6 (b), and the error is acceptable.

Optimized turbocharger-matching of large-displacement PFI hydrogen engine

In order to achieve the desired power at 1800 rpm, the results of the method described above serve as a guide in the turbine

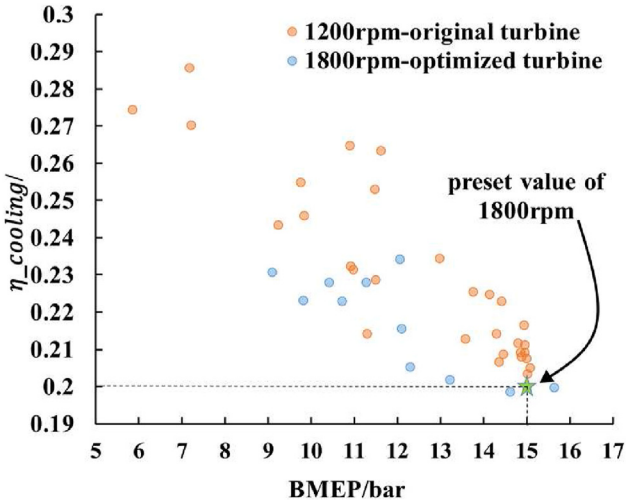


Fig. 7 – The variation of $\eta_{cooling}$ with BMEP @ 1200 rpm and 1800 rpm.

selection process. According to experiments already conducted on the hydrogen engine with the original turbine and required sensors for the temperatures, the value of $\eta_{cooling}$ can be determined. Fig. 7 shows that the $\eta_{cooling}$ at 1200 rpm with high load is 0.21 on average, and a greater engine speed causes a reduction in the $\eta_{cooling}$. Thus, the $\eta_{cooling}$ at 1200 rpm with 1.5 MPa of BMEP is set at 0.20 preliminarily, which needs to be verified according to the results at certain condition. The other parameters are set respectively according to results from previous experiments on the 5.13 L hydrogen engine: the desired power = 120 kW; the lambda = 1.8; compressor pressure ratio = 2.7; brake thermal efficiency = 0.38; air temperature after cooler = 35 °C, turbocharger efficiency = 0.48; turbine discharge pressure = 0.12 MPa due to the high backpressure at high speed.

The turbine inlet temperature and expansion ratio can be calculated as 874.7 K and 2.953 respectively, using the method in Fig. 3 and the formulas above. Furthermore, the reduced mass flow rate are calculated as 0.01417 (kg/s)·K^0.5/kPa. The reduced mass flow rate used as a selection criterion for turbine selection. Finally, an optimized turbine whose maximum reduced mass flow rate is extended to 0.0156 (kg/s)·K^0.5/kPa is selected. The optimized turbocharger is installed on the hydrogen engine and a series of experiments are conducted at 1800 rpm. As the results shown in Fig. 8 (a) and (b), the output power reached 121 kW with a lambda of 1.73 when experiments were conducted with the selected turbine, indicating the performance has been improved further, and the reduced mass flow rate is 0.01394 (kg/s)·K^0.5/kPa. The difference between the calculated and experiment results comes from the waste gate leading to a lower pre-turbine exhaust pressure. The experiment air mass flow is only 548 kg/h much lower than the calculated air mass flow of 601 kg/h due to a lower lambda and lower volumetric efficiency. In addition, Fig. 7 illustrates that the $\eta_{cooling}$ at 1800 rpm is around 0.2 when BMEP is about 1.5 MPa, which is close to the preset value in previous calculation. In order to explore the maximum power of the hydrogen engine, the experiments at 2000 rpm are carried out. The matching points show the expansion ratio and the and reduced mass flow rate increase further at high load, and the power reaches 128.9 kW at 2000 rpm @ 1.71 of the lambda which shows a well-matching. However, further output power is restricted by knock of the hydrogen engine. As shown in Fig. 8(c), the maximum pressure reaches 17.6 MPa 3 cycles after a typical cycle with a maximum pressure of 10.9 MPa due to a sequence of knock occurrences. Additionally, as the cycles go, the rate of heat release increases significantly and the start of combustion is advanced according to Fig. 8(d). The knock must be avoided since it increases friction and cooling loss and can possibly damage H2ICE components.

The following tests results in Fig. 8 (a) and (e) demonstrate the turbocharger matching at other speeds. The hydrogen engine can achieve better power performance at 1400 rpm and 1600 rpm. As compared to the original turbine, the maximum BMEP of 1400 rpm and 1600 rpm are significantly improved, reaching 1.624 MPa and 1.591 MPa respectively. The maximum torque is increased to 662.6 N m at 1400 rpm, and the power is improved to 109 kW at 1600 rpm. On the contrary, the huge mass flow turbine results in lower rotor speed and intake pressure at 1200 rpm where a great torque requirement exists. The expansion ratio remains 1.75, and the torque reaches 517 N m at 1200 rpm @ 1.87 of the lambda, which is a respectable performance but still falls short of the desired torque. The results at 1000 rpm also show a poor matching where the maximum BMEP is only 0.92 MPa. In contrast, the maximum power and torque are improved by 38.7% and 7.7% respectively, however, the maximum torque with the optimized turbine is achieved at 1400 rpm. The torque at 1200 rpm decreases by 15.6%. The performance limitations with the optimized turbine and the original turbine are listed in Table 5.

Furthermore, the exhaust pressures with the original and optimized turbine respectively are contrasted at 1200 rpm and 2000 rpm. The exhaust pressures of the optimized turbine are typically lower than those of the original turbine. As Fig. 9 (a) shows, the exhaust pressure of the original turbine at 1200 rpm grows from 0.197 MPa to 0.262 MPa with the rising BMEP, while that of the optimized turbine increases from 0.138 MPa to 0.196 MPa, which is generally lower. The reason is that the intake pressure of the optimized turbine is low, and more hydrogen is needed to maintain the same BMEP, which indicates a poor charging capacity at low speeds. According to Fig. 9 (b), at 2000 rpm, the exhaust pressure of the original turbine increases from 0.280 MPa–0.320 MPa, and that of the optimized turbine rises from 0.208 MPa to 0.303 MPa. As the BMEP grows, the exhaust pressure difference narrows because the lambda of the original turbine falls obviously to increase the BMEP, which reduces air flow. In contrast, the exhaust pressure of the optimized turbine grows continuously. Excessive exhaust pressure leads to higher pumping loss especially at high speeds. The intake pressure of the optimized turbine reaches the maximum of 0.250 MPa with a BMEP of 1 MPa @ a lambda of 2.15, while that of the original turbine is only 0.22 MPa with a low lambda of 1.69. In contrast, the higher intake pressure brings more air flow and improves the power performance with a higher lambda. Therefore, for the exhaust pressure and the intake pressure at high speed, the optimized turbine is the superior option. According to the analysis, well-matched turbines should have a balanced exhaust pressure and intake pressure at various speeds.

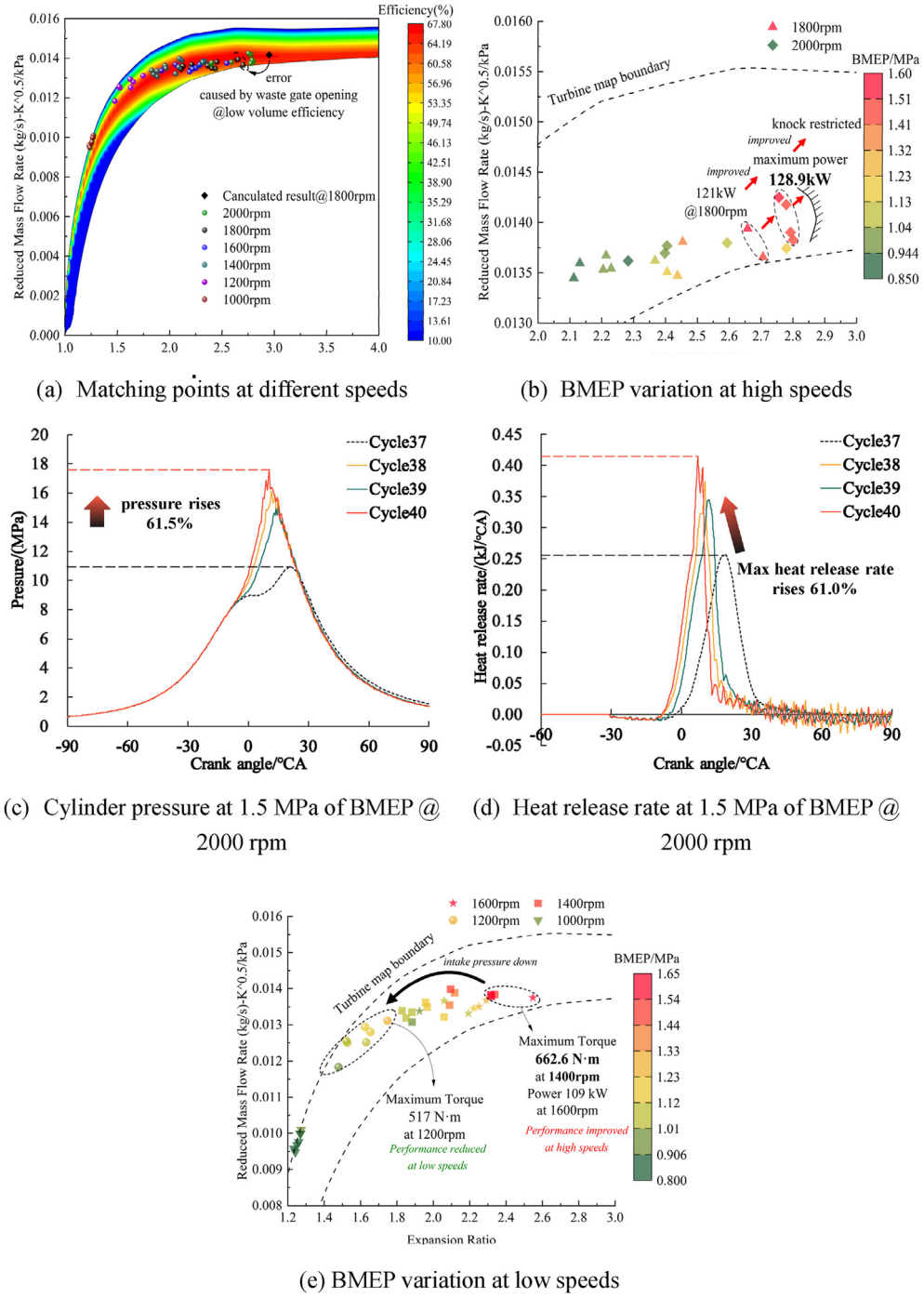


Fig. 8 – (a) The 5.13 L hydrogen engine matching points at different speeds in the optimized turbine efficiency map.

Table 5 – The performance limitations with the optimized turbine and the original turbine.

Performance limitations	Optimized turbine	Original turbine
Most efficient operating range of the engine	1200 rpm–1400 rpm @ 1.1 MPa–1.5 MPa of BMEP	1200 rpm@ 1.1 MPa–1.3 MPa of BMEP
The continuous operating range of the engine	800 rpm–2000 rpm	800 rpm–2000 rpm
The maximum power or torque	128.9 kW @ 2000 rpm	93.0 kW @ 1800 rpm
	662.6 N m @ 1400 rpm	615.0 N m @ 1200 rpm
The lowest fuel consumption range	71.73 g/kWh @ 1400 rpm/1.32 MPa of BMEP	74.21 g/kWh @ 1200 rpm/1.13 MPa of BMEP
The optimum turbocharger efficiency	47.9% @ 1400 rpm/1.32 MPa of BMEP	43.7% @ 1400 rpm/1.13 MPa of BMEP

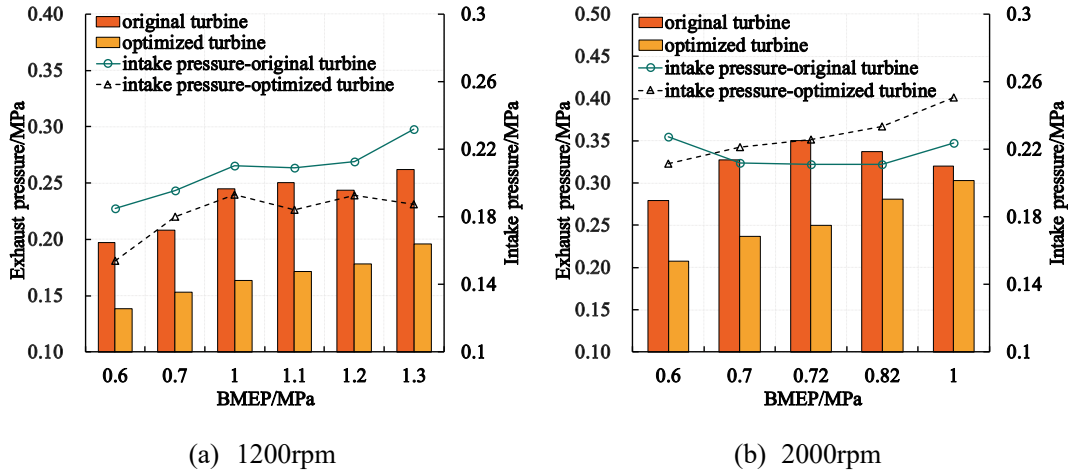


Fig. 9 – The exhaust pressures and intake pressures of the original turbine and the optimized turbine at (a) 1200 rpm and (b) 2000 rpm.

The maximum reduced mass flow rate increases by 61.5% compared the optimized turbine with the original turbine, and the well-matched reduced mass flow rate increases from 0.00904 (kg/s)·K^0.5/kPa at 1200 rpm to 0.01425 (kg/s)·K^0.5/kPa at 2000 rpm, which is a great range for typical waste gate turbocharger. Thus, for a hydrogen engine to reach high power performance at both low and high engine speeds, especially the hydrogen engines whose speed vary significantly, a turbocharger with a wider range of the reduced mass flow rate, like VGT (Variable geometry turbine) is very necessary. The wider flow rate range of VGT provides more potential with H2ICE to adjust the exhaust pressure and intake pressure. The performance of turbocharged H2ICE with VGT and the adjustment will be studied in future researches.

Conclusion

This study presents the process of the optimization of a large-displacement PFI hydrogen engine turbocharger matching for better performance, and a calculation method of turbine selection for turbocharged H2ICEs is utilized. The turbine inlet temperature and expansion ratio are calculated by considering the exhaust components, and the reduced mass flow rate can serve as the reference for turbine selection. Finally, the performance of the large-displacement PFI hydrogen engine is improved significantly. The following conclusions are reached.

1. The desired torque of 615 N m can be achieved at 1200 rpm with the original turbine. However, the original turbine performs poorly at high speeds due to the low reduced mass flow rate of the turbine, whose maximum is 0.0966 (kg/s)·K^0.5/kPa. A modified turbocharger is necessary because of the high air-fuel ratio and the lean burn of hydrogen.
2. The reduced mass flow rate of turbine can be calculated to select a well-matched turbine for H2ICE. Due to different exhaust components and temperatures, the *Cp* of the exhaust gas from H2ICEs is very different from that of typical gasoline engines. The experiment data of the 2.3 L and 5.13 L turbocharged H2ICE is used to validate the calculation method at various engine speeds and throttle openings, and the method is applicable for 5.13 L H2ICE according to the validation. The error of calculated and measured data can be less than 5% mostly.
3. The optimized turbocharger is selected according to the calculated reduced mass flow rate for the 5.13 L PFI hydrogen engine. The power performance at 1800 rpm is improved from to 93 kW–121 kW. The maximum power is extended to 128.9 kW at 2000 rpm, which is increased by 38.7%. In addition, the maximum torque is increased to 662.6 N m with at 1400 rpm, which is increased by 7.7% compared with the maximum torque with original turbine. However, the power performance is reduced by 15.6% at 1200 rpm, and the torque at 1200 rpm decreases to 518 N m. According to analysis, the well-matched turbine should be balanced the exhaust pressure and intake pressure at different speeds.
4. The maximum reduced mass flow rate of the optimized turbine increases by 61.5% comparing with the original turbine, and the variations of the reduced mass flow rates between low and high engine speeds vary significantly from 0.00904 (kg/s)·K^0.5/kPa at 1200 rpm to 0.01425 (kg/s)·K^0.5/kPa at 2000 rpm. Thus the VGT (Variable geometry turbine), which provides a wide flow rate range, may be a better option for hydrogen engines to accommodate different engine speeds. The VGT is a key factor for H2ICEs to achieve high power at different speeds. The characteristics of turbocharged H2ICE with VGT and the adjustment will be studied in future research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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